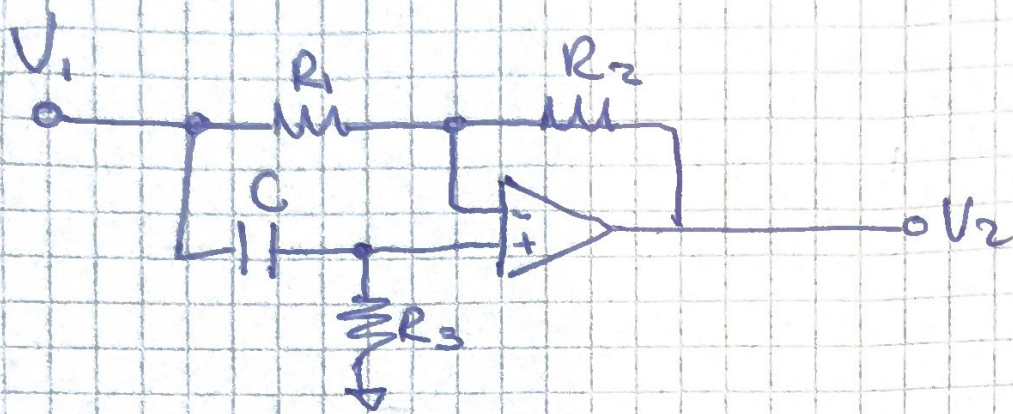
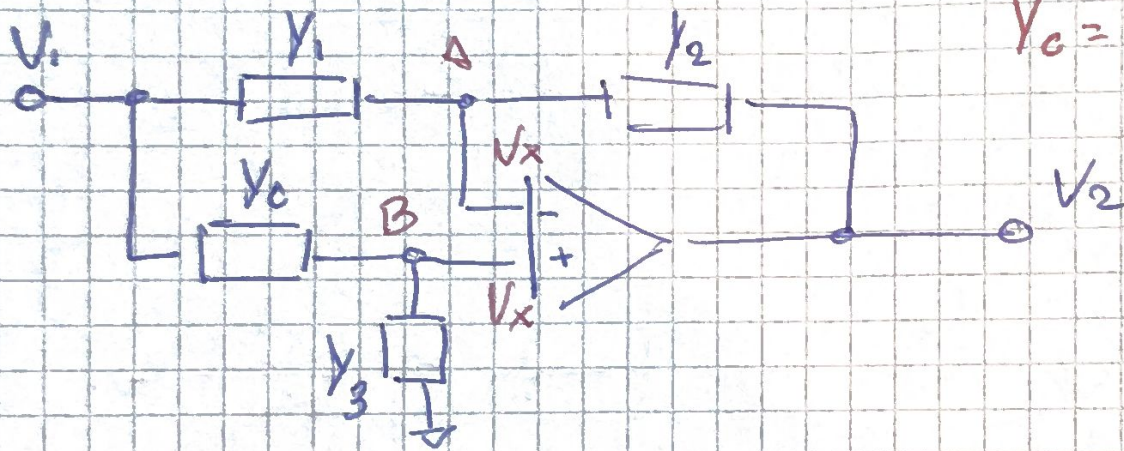




$$Y_1 = \frac{1}{R_1} \quad Y_2 = \frac{1}{R_2}$$

$$Y_C = sC \quad Y_3 = \frac{1}{R_3}$$



Por nodos

$$(A) \quad V_x (Y_1 + Y_2) - V_1 Y_1 - V_2 Y_2 = 0 \quad (1)$$

$$(B) \quad V_x (Y_C + Y_3) - V_1 Y_C - 0 \cdot Y_3 = 0 \quad (2)$$

$$V_x (Y_C + Y_3) = V_1 Y_C$$

$$V_x = V_1 \frac{Y_C}{Y_C + Y_3} \quad (3)$$

(3) en (1)

$$V_1 \cdot \frac{Y_C (Y_1 + Y_2)}{Y_C + Y_3} - V_1 Y_1 = V_2 Y_2$$

$$\left(\frac{Y_C (V_1 + V_2)}{Y_C + Y_3} - Y_A \right) \frac{1}{V_2} = \frac{V_2}{V_1}$$

$$\frac{\cancel{Y_C V_1} + Y_C V_2 - \cancel{V_1 Y_C} - V_1 Y_3}{(Y_C + Y_3) V_2} = \frac{V_2}{V_1}$$

$$\frac{V_2}{V_1} = \frac{Y_C V_2 - V_1 Y_3}{(Y_C + Y_3) V_2} = \frac{Y_C - Y_3 \frac{V_1}{V_2}}{Y_C + Y_3}$$

Reemplazo

$$\frac{V_2}{V_1} = \frac{\$C - \frac{1}{R_3} \cdot \frac{R_2}{R_1}}{\$C + \frac{1}{R_3}}$$

Donde $\Rightarrow \frac{R_2}{R_1} = 1$.

$$\frac{V_2}{V_1} = \frac{\$ - \frac{1}{R_{3C}}}{\$ + \frac{1}{R_{3C}}}$$

$$\left\{ \begin{array}{l} \text{Polo} = -\frac{1}{R_{3C}} \\ \text{Cero} = \frac{1}{R_{3C}} \\ \omega_0 = \frac{1}{R_{3C}} \end{array} \right.$$

Normalización

$$R_N = \frac{R_3}{\Omega_z} \quad \Rightarrow \quad R_3 = \Omega_z \quad \Rightarrow \quad R_N = 1$$

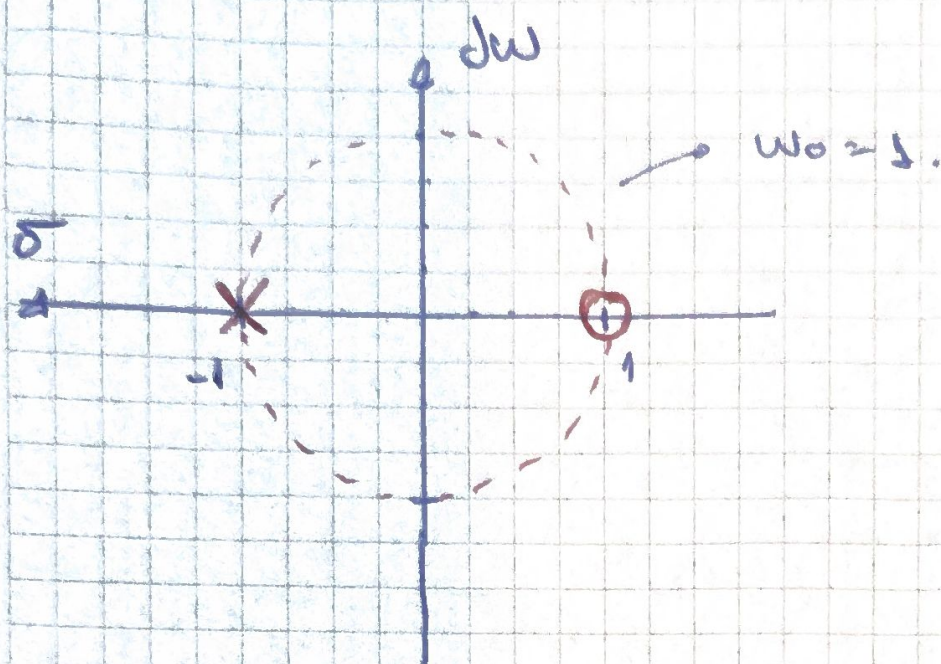
$$\Omega_w = \omega_0 = \frac{1}{R_{3C}}$$

$$C_N = C \cdot \Omega_z \cdot \Omega_w = \cancel{C} \cdot \cancel{R_3} \cdot \frac{1}{\cancel{R_{3C}}} \quad \Rightarrow \quad C_N = 1$$

Transferencia Normalizada

$$T(s) = \frac{\$ - 1}{\$ + 1}$$

Diagrama Polar y Ceros



Fase

$$\varphi = \varphi_z - \varphi_p$$

$$\varphi_z = \pi - \arctan\left(\frac{\omega}{\omega_0}\right)$$

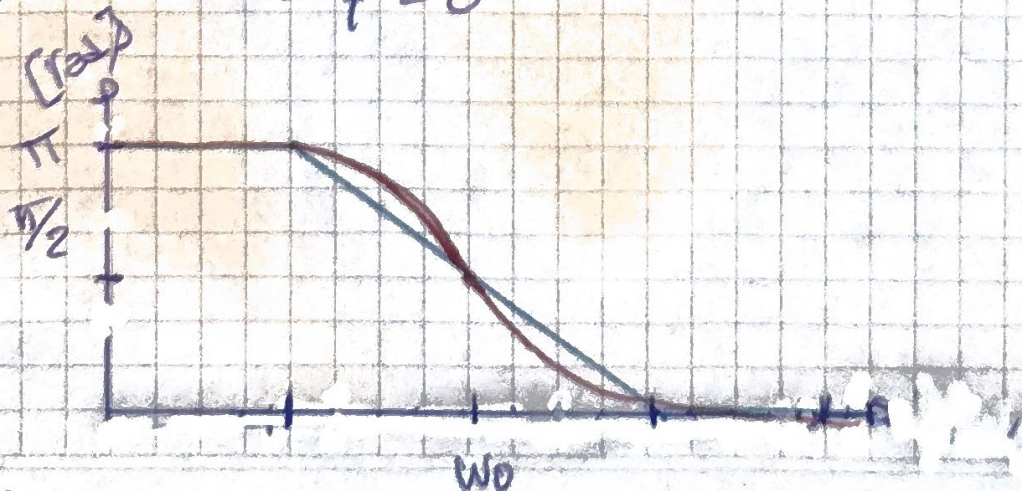
$$\varphi_p = \arctan\left(\frac{\omega}{\omega_0}\right)$$

$$\varphi = \pi - 2 \arctan\left(\frac{\omega}{\omega_0}\right)$$

$$\omega = 0 \Rightarrow \varphi = \pi$$

$$\omega = \omega_0 \Rightarrow \varphi = \pi/2$$

$$\omega \rightarrow \infty \Rightarrow \varphi = 0$$



$$|T(s)| = \frac{\sqrt{(1^2) + (1^2)}}{\sqrt{(1^2) + (1)^2}} = 1.$$

$$20 \text{ dB } \log(1) = 0 \text{ dB}$$

